

MSAJCE — Mohamed Sathak A.J. College of Engineering

Dept. of Computer Science and Business Systems (CSBS) — Complete Data

Extract for RAG Model

Source URL: <https://www.msajce-edu.in/csbs.php> | Sections: 13 sections covered
Scraped: 2026-04-27 | Chunks: 8 | Token Target: 500 | Overlap: 100 | Model: text-embedding-3-small

1. CSBS Department Overview, Key Features & Programme Details

Chunk ID: msajce_csbs_chunk_01 | URL: <https://www.msajce-edu.in/csbs.php#tab1> | ~440 tokens
Context: CSBS overview, TCS industry collaboration, key features, programme details, seats, eligibility and benefits over CSE.

B.Tech in Computer Science and Business Systems (CSBS) is a specialized undergraduate engineering program designed to blend core computer science education with insights into business systems and management. Introduced in collaboration with Tata Consultancy Services (TCS) to bridge the gap between traditional computer science education and the dynamic needs of the IT and business world in the era of Business 4.0.

Curriculum covers: Analytics, Deep Learning, Machine Learning, Cloud Computing, Business Analytics, Financial Management, Economics, Internet of Things, Design Thinking, humanities and value-added science.

Key Features:

- **Balanced Curriculum** — Core CS (Data Structures, Algorithms, OS, DBMS, AI/ML) + Business subjects (Economics, Financial Management, Business Process Management).
- **Industry Collaboration** — Designed in partnership with TCS; emphasis on employability and industry readiness.
- **Focus Areas** — Problem-solving using computational thinking, Business analytics, IT systems for business applications.
- **Soft Skills Training** — Communication, Teamwork, Entrepreneurial thinking.

Benefits Over Regular CSE: More interdisciplinary exposure; better suited for tech-enabled business functions; greater alignment with industry expectations in IT services and consulting.

Detail	Value
Programme	B.Tech. Computer Science and Business Systems (CSBS)
Industry Partner	Tata Consultancy Services (TCS)
Duration	4 years (Regular) / 3 years (Lateral Entry)
Semesters	8 (Regular) / 6 (Lateral Entry)
Total Seats	30
Govt. Quota	15
Mgmt. Quota	15
Eligibility	10+2 with Physics, Chemistry and Mathematics
Curriculum	2021 Regulation — https://www.msajce-edu.in/uploads/academics/2021Regulation.pdf
Higher Studies	M.E. / M.Tech. / M.B.A.
Top Recruiters	TCS, Infosys, Wipro, Accenture, CTS, FSS

Sample Questions this chunk answers:

- Does MSAJCE offer CSBS course?
- What is CSBS at MSAJCE?
- Is CSBS designed with TCS?
- What is the seat intake for CSBS?
- How is CSBS different from regular CSE?

2. Vision, Mission, PEOs, PSOs & All 12 Programme Outcomes (POs)

Chunk ID: msajce_csbs_chunk_02 | URL: <https://www.msajce-edu.in/csbs.php#tab2> | ~480 tokens

Context: CSBS Vision, Mission, PEOs, PSOs and all 12 POs.

Vision

To be a centre of excellence for transforming students into proficient IT Professionals with Business Skills through sustainable practices.

Mission

- M1. Impart core fundamental knowledge and necessary skills in Information Science and Business through innovative teaching and learning methodology.
- M2. Inculcate Business thinking, ethics, lifelong learning and creativity needed for industry and society.
- M3. Cultivate the students with all-round competencies, for career, higher education and self-employability.

Programme Educational Objectives (PEOs)

PEO	Description
PEO1	Graduates prepared for analysing, designing, developing and testing software solutions with creativity and sustainability.
PEO2	Graduates skilled in modern tools for critical problem solving and analysing industrial and societal requirements.
PEO3	Graduates prepared with managerial and leadership skills for career and starting own firms.

Program Specific Outcomes (PSOs)

PSO	Description
PSO1	Develop creative solutions by adapting emerging technologies/tools for real time industry applications.
PSO2	Apply acquired knowledge to develop software solutions and innovative mobile apps for automation applications.

Programme Outcomes (POs) — All 12

PO	Title	Description
PO 1	Engineering Knowledge	Apply knowledge of mathematics, science, engineering fundamentals.
PO 2	Problem Analysis	Identify, formulate and analyze complex engineering problems.
PO 3	Design/Development	Design solutions for complex problems meeting specified needs.
PO 4	Conduct Investigations	Use research-based knowledge and methods for valid conclusions.
PO 5	Modern Tool Usage	Create, select and apply appropriate techniques and modern IT tools.
PO 6	Engineer & Society	Apply reasoning to assess societal, health, safety, legal and cultural issues.
PO 7	Environment & Sustainability	Understand impact of engineering solutions in societal and environmental contexts.
PO 8	Ethics	Apply ethical principles and commit to professional ethics and norms.

PO 9	Team Work	Function effectively as individual, member or leader in diverse and multidisciplinary teams.
PO 10	Communication	Communicate effectively, write reports, make presentations.
PO 11	Project Management	Demonstrate knowledge of engineering and management principles.
PO 12	Life-long Learning	Engage in independent and life-long learning in technological change.

Sample Questions this chunk answers:

- What is the vision of CSBS at MSAJCE?
- What are the PEOs for CSBS?
- What are all 12 programme outcomes for CSBS?
- Does CSBS teach business thinking?

Keywords: CSBS vision, CSBS mission, PEO, PSO, programme outcomes, OBE, business thinking, IT professionals with business skills

3. Employment Growth 2025-2030 — Key Drivers for CSBS Graduates

Chunk ID: msajce_csbs_chunk_03 | URL: <https://www.msajce-edu.in/csbs.php#tab3> | ~460 tokens

Context: Key employment growth drivers for CSBS professionals in India 2025-2030.

Significant employment growth expected driven by digitalization, e-commerce rise, AI adoption, and business process transformation. India is positioning itself as a hub for IT services and business innovation.

Driver	Key Technologies / Areas	Job Impact
Digital Transformation	SAP, Oracle, Microsoft Dynamics, SaaS, ERP	ERP Consultants, Business Systems Analysts
AI, ML and Analytics	Python, Hadoop, Spark, Tableau, Power BI	Data Scientists, BI Analysts, ML Engineers
Cloud Computing	AWS, Azure, Google Cloud, Serverless, multi-cloud	Cloud Architects, Cloud Engineers
E-Commerce Growth	AI chatbots, recommendation systems, supply chain	Data Scientists, Systems Analysts
FinTech Expansion	Blockchain, digital payments, AI financial services	FinTech Professionals, Blockchain Developers
Healthcare Digitalization	Digital records, telemedicine, AI diagnostics	Healthcare IT Professionals
Smart Manufacturing (Industry 4.0)	IoT, AI, cloud in manufacturing	IT-Business Bridge Professionals
Cyber Security & Compliance	Firewalls, encryption, GDPR, Data Protection Bill	Cyber Security Specialists, Compliance Experts

Sample Questions this chunk answers:

- What jobs can CSBS graduates get?
- Is CSBS in demand in India?
- Does CSBS have scope in FinTech?
- Will CSBS graduates get jobs in cybersecurity?

Keywords: CSBS employment, digital transformation, ERP SAP Oracle, cloud computing, FinTech blockchain, cybersecurity, AI ML analytics, e-commerce India, Industry 4.0

4. Key Job Roles in High Demand — CSBS Graduates 2025-2030

Chunk ID: msajce_csbs_chunk_04 | URL: <https://www.msajce-edu.in/csbs.php#tab3> | ~490 tokens

Context: Detailed job roles for CSBS graduates with responsibilities and required skills.

Job Role	Key Skills Required
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Business Systems Analyst	Business analysis, ERP (SAP, Oracle), Agile, Requirements gathering, Systems design
Data Scientist / Data Analyst	Python, R, SQL, ML, Tableau, Power BI, Hadoop, Spark, statistical modeling
Cloud Architect / Cloud Engineer	AWS, Azure, Google Cloud, cloud architecture, Serverless technologies, cloud security
ERP Consultant / Implementation Specialist	SAP, Oracle, business process modeling, project management, change management
AI / ML Engineer	Python, TensorFlow, Keras, AI algorithms, NLP, data preprocessing
Cyber Security Specialist	Network security, ethical hacking, penetration testing, data encryption, GDPR
Business Intelligence Analyst	SQL, Tableau, Power BI, data mining, business analytics, decision support systems
Blockchain Developer	Ethereum, Hyperledger, smart contracts, cryptography, distributed ledgers
FinTech Professional	FinTech platforms, blockchain, digital payments, regulatory compliance, data analytics

Sample Questions this chunk answers:

- What does a Business Systems Analyst do?
- Can CSBS graduates become data scientists?
- What skills are needed for an ERP Consultant?
- What is the role of a Blockchain Developer?

Keywords: business systems analyst, data scientist, cloud architect, ERP consultant SAP Oracle, AI ML engineer, cyber security, BI analyst, blockchain developer Ethereum, FinTech professional

5. Employment Sectors & CSBS Faculty List

Chunk ID: msajce_csbs_chunk_05 | URL: https://www.msajce-edu.in/csbs.php#tab3 | ~470 tokens

Context: Sectors with highest employment potential for CSBS graduates and complete faculty list.

Sector	Key Opportunities
Information Technology (IT Services)	Cloud computing, BPO, ERP implementation
E-Commerce	Data analytics, customer experience design, supply chain management
Financial Services and FinTech	Digital payments, blockchain, insurtech, investment technology
Healthcare	Digital health, telemedicine, AI for diagnostics
Manufacturing and Industry 4.0	IoT, AI, cloud computing integration with industrial processes

CSBS Department Faculty

Name	Qualification	Designation	Specialization
Dr. NAGASUBRAMANIAN R	M.E., Ph.D.	Head — PG	Computer Science Engineering
Dr. SUBRAMANIAN P	M.E., Ph.D.	Head — UG	Wireless Sensor Networking & Image Processing
Dr. PREMALATHA G	M.E., Ph.D.	Associate Professor	Image Processing and Machine Learning
Mrs. SARANYA V	M.Tech., (Ph.D)	Asst. Professor	IOT and Cloud Computing
Mrs. SASIKALA L	M.E.	Asst. Professor	Computer Science and Engineering
Mrs. KIRUTHIKA S	M.E.	Asst. Professor	Computer Science and Engineering
Mrs. ASRIN MAHMOOTHA A	M.E.	Asst. Professor	Computer Science Engineering
Mr. PRAVEENKUMAR P	M.E.	Asst. Professor	Computer Science Engineering

Mr. RAJAKUMAR B	M.Tech., (Ph.D)	Asst. Professor	IOT, AI, NLP and Image Processing
Mr. RAJASEKAR R	M.E., (Ph.D)	Asst. Professor	IOT, Network Security

Sample Questions this chunk answers:

- Who is the HOD of CSBS at MSAJCE?
- Who are the CSBS faculty?
- What sectors have the most jobs for CSBS graduates?

Keywords: CSBS HOD, Dr Nagasubramanian, Dr Subramanian, Dr Premalatha, CSBS faculty, IOT Cloud Computing, image processing, NLP, IT services, FinTech, Industry 4.0

6. CSBS Laboratories — 6 Labs with Resource Links

Chunk ID: msajce_csbs_chunk_06 | URL: <https://www.msajce-edu.in/csbs.php#tab6> | ~420 tokens

Context: All 6 CSBS department laboratories with course codes, names, and virtual lab links.

No	Lab Name	Code	Course Name	Virtual Lab URL
1	C Programming Lab	CS8261	C Programming	https://www.msajce-edu.in/images/departments/csbs/Virtual-Lab-C-Programming.pdf
2	Internet Programming Lab	CS8661	Internet Programming	https://www.msajce-edu.in/images/departments/csbs/Virtual-Lab-Internet-Programming.pdf
3	Operating Systems Lab	CS8461	Operating Systems	https://www.msajce-edu.in/images/departments/csbs/Virtual-Lab-OS.pdf
4	OOAD Lab	CS8461	Object Oriented Analysis and Design	https://www.msajce-edu.in/images/departments/csbs/Virtual-Lab-ood.pdf
5	Web Technology Lab	IT8511	Web Technology	https://www.msajce-edu.in/images/departments/csbs/Virtual-Lab-Web-Technology.pdf
6	Networks Lab	CS8581	Networks Lab	https://www.msajce-edu.in/images/departments/csbs/Networks.pdf

Sample Questions this chunk answers:

- What labs are available in CSBS at MSAJCE?
- Does CSBS have a Networks lab?
- Does CSBS have a Web Technology lab?
- What is the course code for C Programming in CSBS?

Keywords: CSBS labs, C programming lab CS8261, internet programming CS8661, operating systems CS8461, OOAD lab, web technology IT8511, networks lab CS8581, virtual lab CSBS

7. Course Materials — 2021 Regulation (9 Subjects)

Chunk ID: msajce_csbs_chunk_07 | URL: <https://www.msajce-edu.in/csbs.php#tab6> | ~455 tokens

Context: All 9 CSBS subjects under 2021 Regulation with codes and resource download links.

No	Code	Subject Name	Lesson Plan	Qn. Bank	Lecture Notes
1	MA3354	Discrete Mathematics	LP	QB	LN
2	CS3351	Digital Principles and Computer Organization	LP	QB	LN
3	AD3391	Database Design and Management	LP	QB	LN
4	AD3351	Design and Analysis of Algorithms	LP	QB	LN
5	AD3301	Data Exploration and Visualization	LP	QB	LN

6	AL3391	Artificial Intelligence	LP	QB	LN
7	AD3381	Database Design and Management Laboratory	LP	QB	LN
8	AD3311	Artificial Intelligence Laboratory	LP	QB	LN
9	GE3361	Professional Development	LP	QB	LN

Note: All PDFs at base URL <https://www.msajce-edu.in/academics/csbs/2021/> with subfolders LessonPlan/, QuestionBank/, LectureNote/, ICTTools/

Sample Questions this chunk answers:

- Where can I get CSBS lecture notes?
- What subjects are in CSBS 2021 regulation?
- Where to find CSBS question banks?
- What is the subject code for Artificial Intelligence in CSBS?

Keywords: CSBS course materials, 2021 regulation, MA3354, CS3351, AD3391, AD3351, AD3301, AL3391, AD3381, AD3311, GE3361, lesson plan, question bank, lecture notes

8. CSBS Placement Companies, Key Links & General Contact

Chunk ID: msajce_csbs_chunk_08 | URL: <https://www.msajce-edu.in/csbs.php> | ~400 tokens

Context: All companies recruiting CSBS graduates from MSAJCE, key download links, and general contact.

S.No.	Company (Group 1)	Company (Group 2)
1–4	Accenture	AT&T
5–8	BYJU'S	Capgemini
9–12	Carbonite	Chella Software
13–16	CISCO	City West Water
17–20	Cognizant Technology Solutions (CTS)	Deloitte Consulting
21–24	Dell Technologies	EMIS Health
25–28	Future Focus Infotech	HCL Technologies
29–32	HP (Hewlett Packard)	HTC
33–36	IBM	Infosys
37–40	Infoview Technologies	Intellect Design Arena
41–44	L&T Infotech	Newage Software
45–48	Oracle	Saudi Airlines
49–52	Solverminds	Sutherland
53–56	TCS (Tata Consultancy Services)	Tech Mahindra
57–60	Wells Fargo	Wipro
61	Wolters Kluwer ELM	—

Sample Questions this chunk answers:

- Which companies recruit from MSAJCE CSBS?
- Does TCS recruit from MSAJCE CSBS?
- Does Deloitte hire CSBS graduates from MSAJCE?
- What is the MSAJCE contact number?

Keywords: CSBS placement, TCS, Wipro, Infosys, HCL, IBM, Accenture, Capgemini, Deloitte, CISCO, Oracle, Wells Fargo, off-campus placement, 99400 04500

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